Electrical Engineering Department
National Institute of Technology Kurukshetra

Subject: Power Electronics-1
Mid. Sem. - I
Note: Attempt all questions.

Time: 50 Minutes
Max. Marks: 15

Q1. Describe the construction and working principle of the IGBT. Also, write the advantages of IGBT over other power electronics switches with proper reason.

Q2. Demonstrate the various firing circuits used to turn on SCR in detail with all the relevant diagrams/waveforms.

Q3. Three SCRs are connected together to form a series string. The voltages across the thyristors are 350V, 300V and 250V respectively. What will be the value of equalising resistance to be used across each thyristor?